AN EXPLICIT SEPARATION OF THE TYPE AND TEST MODEL STRUCTURES ON DE MORGAN CUBICAL SETS, AND THEIR DIFFERENT CONTRACTIBLE OBJECTS

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ABSTRACT. On presheaves over the De Morgan cube category — the site of CCHM cubical type theory — the weak equivalences $\mathsf{W}_{\mathrm{type}}$ of the cubical-type model structure and the weak equivalences $\mathsf{W}_{\mathrm{test}}$ of Cisinski's test structure satisfy $\mathsf{W}_{\mathrm{type}} \subseteq \mathsf{W}_{\mathrm{test}}$; the expected non-coincidence has circulated since a 2018 folklore argument of Sattler and Coquand and is recorded in recent published work. This paper proves two theorems. First, an explicit direct separation: a concrete map $\bar{\Phi} \in \mathsf{W}_{\mathrm{test}} \setminus \mathsf{W}_{\mathrm{type}}$, from a Klein quotient of the join $\square^1 \star \square^1$ to the Klein quotient of the square, built from the De Morgan median interpolation. On the test side both objects realize $B\mathbb{Z}/2 \star B\mathbb{Z}/2$ and $\bar{\Phi}$ is an equivalence; on the type side we prove a nullity theorem — every map from $\square^2/K$ into a fibrant replacement of the join model is null-homotopic — via a strictification lemma, two three-line parity arguments in free De Morgan algebras, and an equivariant slide along uniform Kan filling cylinders. Second, an object-level separation: a subdivided mapping cone of the same witness is an explicit finite presheaf which is test-contractible but not type-contractible. The two structures therefore do not even agree on which objects are aspherical, and separation occurs at the terminal representable. The second theorem is reduced to the first by total sector-projections on the subdivided cone, with no case analysis on how invariant classes are twisted over the quotient; a boundary-constrained satisfiability method makes the relevant level-three cells finitely searchable. All finite combinatorial claims are machine-verified.

1. INTRODUCTION

Let $\square_{\mathrm{DM}}$ be the De Morgan cube category and $\widehat{\square_{\mathrm{DM}}}$ its presheaf topos, the site of CCHM cubical type theory [6]. Two homotopy theories live on $\widehat{\square_{\mathrm{DM}}}$: the *cubical-type model structure*, whose fibrations are the uniform Kan fibrations of type theory, and the *test model structure* of Cisinski [7], whose weak equivalences are created by the category of elements. The containment

$$\mathsf{W}_{\mathrm{type}} \subseteq \mathsf{W}_{\mathrm{test}} \tag{U}$$

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holds on any totally aspherical test cube category ([15]; reproved here as Proposition 2.3). That it is strict — that the *standard* cubical-type model structure on De Morgan cubical sets does not coincide with the spatial (test) homotopy theory — has been expected since a 2018 folklore argument of Sattler and Coquand [13], an expectation recorded in the introduction of the recent landmark [3]; and [15] showed that over the square no *object-level* witness of the expected kind exists: every isotropy quotient of the square has the same weak-contractibility status in the two structures.

This paper proves two theorems: the containment (U) is strict, by an explicit map; and — what the object-level analysis of [15] makes surprising — the difference is nevertheless visible on objects, at the price of leaving the class of isotropy quotients. In one sentence: *the two natural homotopy theories carried by the site of cubical type theory disagree already about which finite objects are contractible.*

Theorem 1.1 (Map-level separation; Theorem 7.1)**.** *There is an explicit map $\bar{\Phi} \in \mathsf{W}_{\text{test}} \setminus \mathsf{W}_{\text{type}}$: the median collage from the Klein quotient of the join $\square^1 \star \square^1$ to the Klein quotient $W = \square^2/K$ of the square. In particular $\mathsf{W}_{\text{type}} \subsetneq \mathsf{W}_{\text{test}}$.*

Theorem 1.2 (Object-level separation; Theorem 12.6)**.** *There is an explicit finite presheaf C — a subdivided mapping cone of $\bar{\Phi}$ — which is test-contractible and not type-contractible. Consequently the two structures have different classes of aspherical objects, and the map $C \to \square^0$ lies in $\mathsf{W}_{\text{test}} \setminus \mathsf{W}_{\text{type}}$: separation occurs at the terminal representable.*

To our knowledge, Theorem 1.1 provides the first explicit direct proof of the folklore expectation in the literature, in a strong form: not an abstract non-coincidence but a concrete separating map, whose test-side equivalence and type-side non-equivalence are each established by finite, machine-checkable computations wrapped in structural arguments. The second theorem answers the question that the first leaves open. A separating *map* is compatible, in principle, with the two structures agreeing about every *object*; and indeed on all isotropy quotients of the square they do agree [15]. Theorem 1.2 says the agreement stops there: a finite complex built from the witness itself is trivial for one structure and not the other. For the semantics of cubical type theory the two theorems together sharpen the folklore in a way we find striking: the type-theoretic homotopy theory of De Morgan cubical sets not only differs from the homotopy theory of spaces — it disagrees with it about which finite objects are trivial.

The witness. Both theorems rest on one construction. The De Morgan median interpolation $F(d,a,t) = (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a)$ assembles into the *median collage* $\Phi = (F(d,a,t), F(d,\neg a,t)) \colon J \to \square^2$ from the join model J, equivariantly for the Klein group, and descends to $\bar{\Phi} \colon J/K \to W$. The test side of Theorem 1.1 identifies both categories of elements with the join $B\mathbb{Z}/2 \star B\mathbb{Z}/2$ and $\mathrm{el}(\bar{\Phi})$ with an equivalence (Theorem 4.1). The type side is a nullity theorem (Theorem 6.3): *every* map from W into a fibrant replacement of J/K is null-homotopic, so no map — in particular no homotopy inverse of $\bar{\Phi}$ — can hit the unit, and $\bar{\Phi} \notin \mathsf{W}_{\text{type}}$ follows since W is not type-contractible (Corollary 2.4). The proofs of these statements occupy Sections 3–7 and are those of the separation part of this series; the parity lemmas at their core are three-line arguments in free De Morgan algebras, and every finite claim is verified by exhaustive computation.

From the map to an object. Theorem 1.2 is proved in Sections 8–12. The witness is the *subdivided* mapping cone

$$C \;=\; \Big(\square^2 \,\cup_\Phi\, (J \times \square^1) \,\cup_J\, \operatorname{cone}(J)\Big)/K :$$

gluing the mapping cylinder of Φ to the cone on a *second* cylinder copy of J before collapsing, and only then dividing by the Klein action. Test-contractibility is a homotopy-pushout computation from $\bar{\Phi} \in \mathsf{W}_{\text{test}}$. For type-non-contractibility, suppose a contraction existed; by the strictification lemma it is a chain of invariant homotopies from the tautological class ι to the vertex constant, analyzed in the equivariant replacement of the cover. The subdivision buys the key geometric fact (Lemma 10.2): no cell touches both the base $\square^2$ and the cone vertex, so the replacement splits into two closed sectors — a cylinder sector carrying a *totally defined* strict retraction $\tilde{r}$ to $\square^2$, and a cone sector carrying a totally defined slide to the vertex — with interface exactly a middle copy of J. A first-exit analysis (Lemma 12.4) shows any chain must present an honest equivariant cell j_0 of the interface, and projecting the chain by $\tilde{r}$ exhibits $\mathrm{id}_W \simeq R(\bar{\Phi}) \circ \pi(j_0)$ — exactly the configuration that the nullity theorem forbids. No case analysis on the twists of invariant classes over the quotient is needed: the sector projections apply to all of them alike. The scheme:

$$R_K(\square^2) \xleftarrow{\;\tilde{r}\;} R_K(\widetilde{M}) \hookrightarrow R_K(\widetilde{C}) \hookleftarrow R_K(\operatorname{cone} J) \xrightarrow{\text{slide}} *$$

Why the object-level question is delicate. Two natural strategies fail, and their failure shaped the construction. The strata circles of W give two canonical loops, strictly rigid at stage zero — no strict disc bounds them — yet a single invariant square homotopes each of them, freely, onto a constant at the *other* vertex (Section 8): the fundamental-group theory of W shows no type/test difference, and free slides across the quotient dissolve based rigidity. A monodromy functional from the branched double covers of W fails as well, exactly at the branch loci. The subdivided cone is engineered so that the analogous slides can only *help*: they normalize the contraction into the cone sector instead of dissolving the obstruction.

Relation to the literature. The comparison problem and its history — the folklore argument, the equivariant repair in the cartesian setting, and the object-level agreement over the square — are discussed with full references in Section 14. Here we note only the localizer-theoretic reading of Theorem 1.2:

for localizers on a test category in the sense of [7], it exhibits two naturally occurring localizers distinguished by their classes of aspherical objects, with an explicit finite witness; and the type-theoretic reading: the finite object C is a concrete complex on which the CCHM fibrant replacement and the spatial homotopy type disagree about triviality itself. The classification of all isotropy quotients of dimension ≤ 3, which builds on the present results, appears in the companion paper [16].

Outline. Section 2: preliminaries, the descent lemma, the algebraic replacement, and (W1). Sections 3–7: the join model, the median collage, the test equivalence, strictification and the parity lemmas, the nullity theorem, and Theorem 7.1. Section 8: the fundamental group of W sees no difference. Section 9: level-three cells as a satisfiability problem, and the stage-zero computations. Section 10: the subdivided cone and its test-contractibility. Section 11: strictification, sectors, and the two total projections. Section 12: the first-exit analysis and Theorem 12.6. Section 13: machine verification. Section 14: related work and outlook.

2. Preliminaries

2.1. **The De Morgan cube category and its presheaves.** We write $\mathrm{DM}(i_1,\dots,i_m)$ for the free De Morgan algebra on generators $i_1,\dots,i_m$: the free bounded distributive lattice with an involution $\neg$ satisfying the De Morgan laws, *without* excluded middle. Its elements have canonical disjunctive normal forms (antichains of clauses over the literals) [6]. $\square_{\mathrm{DM}}$ has objects $[m]$, $m \geq 0$, and $\square_{\mathrm{DM}}([m],[n]) := \mathrm{DM}(i_1,\dots,i_m)^n$, with composition by substitution. We write $\widehat{\square_{\mathrm{DM}}}$ for the presheaf category, $\square^n$ for the representable on $[n]$, and use x, y (resp. d, a, t) for the coordinates of $\square^2$ (resp. $\square^3$).

We will use the representation of the free algebra by monotone functions: DM_n is isomorphic to the lattice of monotone maps $\{0,1\}^{2n} \to \{0,1\}$ on the *literal cube* L_n, with coordinates in pairs $(x_i, \neg x_i)$; the involution is

$$(\neg\varphi)(p) \;=\; 1 - \varphi(\rho p), \tag{1}$$

where ρ swaps each literal pair and complements it [14]. Two total involutions of the literal cube will appear below: the *generator swap* sw (exchange the x-pair with the y-pair) and the *literal swap* composed with ρ, which is the bitwise complement $c(p) = \bar{p}$.

2.2. **The two structures.** The *type structure* on $\widehat{\square_{\mathrm{DM}}}$ is the model structure underlying the CCHM interpretation: cofibrations are the monomorphisms, fibrations the *uniform* Kan fibrations — maps equipped with fillers for the open boxes $(m \mathbin{\hat{\otimes}} \delta^{\varepsilon})$, m a monomorphism, chosen strictly naturally in pullback along substitutions — and weak equivalences $\mathsf{W}_{\mathrm{type}}$; see [6] for the type theory, [8] for uniform fibrations, and [12] for the model structure. Every object is cofibrant.

Two structural facts about fibrant replacement are used; both hinge on *uniformity.* Recall that in any presheaf category the monomorphisms are cellularly generated by a set of inclusions with quotients of representables as codomains (existence of cellular models, Cisinski [7, §1.2]); on a site with diagonals, connections and reversals — which is not (generalized) Reedy, cf. the discussion in [4] — the quotient shapes are genuinely needed, and representable boundaries alone do not suffice. Uniformity is what nevertheless confines fibrant replacement to representable-shaped data.

Let $\mathbb{J}$ denote the small category whose objects are the box inclusions $j_{S,n,\varepsilon} := (S \hookrightarrow \square^n)\hat{\otimes}\delta^\varepsilon$ of Lemma 2.1 below, and whose morphisms $j_{S',m,\varepsilon} \to j_{S,n,\varepsilon}$ are the substitutions $\sigma \in \square_{\mathrm{DM}}([m],[n])$ carrying S' into S; such a σ induces the box morphism $\hat{\sigma} := \sigma \times \mathrm{id}_{\square^1}$. A *box* for $j = j_{S,n,\varepsilon}$ over X is a map $b : \mathrm{dom}(j) = S \times \square^1 \cup \square^n \times \{\varepsilon\} \to X$; a morphism σ restricts boxes, $b \mapsto b \circ \hat{\sigma}$. Uniform fibrancy is precisely a right lifting structure against the *category* $\mathbb{J}$ (fillers for all boxes, compatible with restriction), the formulation of [8].

Lemma 2.1 (Descent via uniformity)**.** *An object equipped with fillers for the boxes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, where S ranges over arbitrary subpresheaves of representables, strictly natural in substitution, is fibrant.*

Proof. Let W carry the stated filling structure; we show that $W \to 1$ has the right lifting property against $m \hat{\otimes} \delta^\varepsilon$ for every monomorphism $m : X \hookrightarrow Y$.

Step 1: cellular decomposition of m. Well-order the nondegenerate cells of Y not in X, say $(y_\beta)_{\beta<\lambda}$ with $y_\beta \in Y([n_\beta])$, and let $X_\beta \subseteq Y$ be the subpresheaf generated by X together with $\{y_\gamma : \gamma < \beta\}$; then $X_0 = X$, $X_\lambda = Y$, and $X_{\beta+1} = X_\beta \cup R_\beta$, where $R_\beta \subseteq Y$ is the image of $y_\beta : \square^{n_\beta} \to Y$. Writing $p_\beta : \square^{n_\beta} \twoheadrightarrow Q_\beta := \square^{n_\beta}/E_\beta$ for the image factorization of y_β — so E_β is the kernel congruence, $p_\beta\sigma = p_\beta\tau \iff y_\beta \cdot \sigma = y_\beta \cdot \tau$ — the square

$$X_{\beta+1} \;=\; X_\beta \cup_{P_\beta} Q_\beta, \qquad P_\beta := Q_\beta \cap X_\beta,$$

is a pushout (unions of subpresheaves are effective pushouts over the intersection in a presheaf topos). On this non-Reedy site the cell shapes Q_β are genuinely quotients of representables [4], which is why representable *boundaries* alone would not suffice; this is the cellular-model phenomenon of [7, §1.2].

Step 2: reduction to one attachment. A lift against $m \hat{\otimes} \delta^\varepsilon$ is an extension of a given $h : Y \times \{\varepsilon\} \cup_{X\times\{\varepsilon\}} X \times \square^1 \to W$ to $Y \times \square^1$. We construct compatible extensions $L_\beta : Y \times \{\varepsilon\} \cup X_\beta \times \square^1 \to W$ by transfinite induction, with $L_0 = h$ and unions at limit stages (the value of L on each cell stabilizes, so the union is well-defined). At a successor stage the required datum is an extension of

$$b_\beta : Q_\beta \times \{\varepsilon\} \,\cup\, P_\beta \times \square^1 \longrightarrow W$$

(restriction of L_β and h) to $Q_\beta \times \square^1$: a filling problem for the box $(P_\beta \hookrightarrow Q_\beta) \hat{\otimes} \delta^\varepsilon$.

Step 3: pullback to the representable and descent. Let $S := p_\beta^{-1}(P_\beta) \subseteq \square^{n_\beta}$ and let $b := b_\beta \circ (p_\beta \times \mathrm{id})$ be the pulled-back box for $j_{S,n_\beta,\varepsilon} = (S \hookrightarrow \square^{n_\beta}) \hat{\otimes} \delta^\varepsilon$; the filling structure provides $\mathrm{val}(b) : \square^{n_\beta} \times \square^1 \to W$ extending b. Now let $\sigma, \tau \in \square_{\mathrm{DM}}([m],[n_\beta])$ be any congruence pair, $p_\beta\sigma = p_\beta\tau$. Then $\sigma^{-1}(S) = (p_\beta\sigma)^{-1}(P_\beta) = (p_\beta\tau)^{-1}(P_\beta) = \tau^{-1}(S) =: S'$, and the restricted boxes agree, $b \circ \hat{\sigma} = b \circ \hat{\tau}$ as boxes for $j_{S',m,\varepsilon}$, because b factors through $Q_\beta \times \square^1$. Strict naturality gives

$$\mathrm{val}(b) \circ \hat{\sigma} \;=\; \mathrm{val}(b \circ \hat{\sigma}) \;=\; \mathrm{val}(b \circ \hat{\tau}) \;=\; \mathrm{val}(b) \circ \hat{\tau}.$$

Since Q_β is the coequalizer of its congruence pairs and $(-)\times\square^1$ preserves this coequalizer (quotients and products of presheaves are computed levelwise), $\mathrm{val}(b)$ descends to $\bar{v} : Q_\beta \times \square^1 \to W$. On the box domain, $\bar{v}$ agrees with b_β because $\mathrm{val}(b)$ extends b and $p_\beta \times \mathrm{id}$ maps $S \times \square^1 \cup \square^{n_\beta} \times \{\varepsilon\}$ onto $P_\beta \times \square^1 \cup Q_\beta \times \{\varepsilon\}$. Set $L_{\beta+1} := L_\beta \cup \bar{v}$; this is well-defined precisely because $\bar{v}$ restricts to L_β on the overlap $P_\beta \times \square^1$.

The induction terminates at λ with the required lift $L_\lambda : Y \times \square^1 \to W$. $\square$

Lemma 2.2 (Algebraic fibrant replacement)**.** *Every $X \in \widehat{\square_{\mathrm{DM}}}$ admits a functorial trivial cofibration $u : X \to R$ with R uniformly fibrant, constructed as the colimit of a transfinite sequence $X = R_0 \to R_1 \to \cdots$ of monomorphisms in which*

$$R_{\beta+1} \;:=\; \mathrm{coeq}\Big(\coprod_{\substack{\sigma: j' \to j \\ b \text{ box for } j \text{ over } R_\beta}} \mathrm{cod}(j') \;\rightrightarrows\; \coprod_{(j,b)} \mathrm{cod}(j) \Big) \sqcup_{\coprod_{(j,b)} \mathrm{dom}(j)} R_\beta,$$

the two parallel arrows being the insertion at $(j', b \circ \hat{\sigma})$ and the composite of $\hat{\sigma}$ with the insertion at (j,b), and the pushout taken along the evaluations b. Consequently:

(1) *every cell of $R_{\beta+1}$ is a cell of R_β or a* filler value $\mathrm{val}(b,c)$*, $c \in \mathrm{cod}(j)$, and the identifications among filler values are* exactly *those generated by*
   (i) *$c \in \mathrm{dom}(j)$:* $\mathrm{val}(b,c) = b(c)$*, a cell of R_β (the pushout);*
   (ii) $\mathrm{val}(b \circ \hat{\sigma}, c) = \mathrm{val}(b, \hat{\sigma} \circ c)$ *(the coequalizer);*

(2) *the stage maps are monomorphisms and no identifications are imposed on old cells: the evaluations agree on the cells identified by the coequalizer, precisely because $b \circ \hat{\sigma}$ is by definition the restricted box; hence equality of cells of R is detected at a stage and is generated by (i)–(ii);*

(3) *filler values are strictly natural,* $\mathrm{val}(b,c) \cdot f = \mathrm{val}(b, c \cdot f)$*, and R carries the tautological uniform filling structure, so R is fibrant by Lemma 2.1;*

(4) *by the standard property of the algebraically free awfs generated by $\mathbb{J}$, the unit u lies in the left class cofibrantly generated by the box inclusions — Garner's construction interposes coequalizers, so u is a retract of a transfinite composite of pushouts of the generators*

rather than literally such a composite — and since the box inclusions are trivial cofibrations of the type structure, whose left class is closed under pushout, transfinite composition and retract, u is a trivial cofibration; the construction is functorial in X.

Proof. This is Garner's algebraic small object argument [9] for the generating *category* $\mathbb{J}$ (rather than a mere set), in the free-monad-on-a-pointed-endofunctor form; the displayed one-step construction is its explicit colimit presentation, as in the treatment of uniform fibrations in [8]. The listed properties are read off from the presentation: (1) is the description of a coequalizer-then-pushout of coproducts of presheaves, whose cell-level relations are precisely the displayed generators; for (2), both relation schemes have a fresh member on each side except (i), whose old member is uniquely determined, and chains through fresh cells terminating in old cells on both ends force equal old cells because restricted boxes evaluate compatibly ($b(\hat{\sigma} \circ c) = (b \circ \hat{\sigma})(c)$ on the box domain); (3) the presheaf structure on a coproduct of representable-products acts on the c-coordinate, and the filling structure assigns to a box its inserted values, uniformity holding by (ii); (4) is the standard comparison of the algebraically free awfs with the cofibrantly generated one [9]. $\square$

The *test structure*: $\square_{\mathrm{DM}}$ is a strict test category (the cube category with connections is one by [11]; the De Morgan case is covered by the criteria of [7], cf. [2]), so the functor $X \mapsto N(\mathrm{el}(X))$ detects a class $\mathsf{W}_{\mathrm{test}}$ of weak equivalences making $\widehat{\square_{\mathrm{DM}}}$ a model for the homotopy theory of spaces. We freely write $\mathrm{el}(X) \simeq Z$ for "the nerve of the category of elements of X is weakly homotopy equivalent to Z".

The following statements, first assembled in [15], will be used. (U) is proved below as Proposition 2.3, (W2) is reproved in Appendix A, and the part of (L) entering the main proofs is reproved there as well. The remaining item (Collage) is not a result of [15] but the standard collage/homotopy-colimit formula, which we use directly from Bousfield–Kan [1] and Thomason [18]; the proofs of Theorems 7.1 and 12.6 therefore do not depend on [15].

(U) $\mathsf{W}_{\mathrm{type}} \subseteq \mathsf{W}_{\mathrm{test}}$.

(W2) Let $K = \{1, \mathrm{sw}, \mathrm{nb}, g\}$ act on $\square^2$ by $\mathrm{sw}(x, y) = (y, x)$, $\mathrm{nb}(x, y) = (\neg x, \neg y)$, $g = \mathrm{nb} \circ \mathrm{sw}$, and let $W := \square^2/K$ (levelwise orbits). Then $\mathrm{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$: the category of elements decomposes into the stabilized sieve (the diagonal and antidiagonal layers, each $\simeq B\mathbb{Z}/2$) and the free cosieve ($\simeq BK$), and the collage evaluates to the join. In particular $\mathrm{el}(W)$ is simply connected with $H_3(\mathrm{el}W; \mathbb{Z}) = \mathbb{Z}/2$; it is not weakly contractible. (For self-containedness, a complete proof of (W2) from (L) and (Collage) is given in Appendix A.)

(L) $\mathrm{L} := \square^1/\langle\neg\rangle$ satisfies $\mathrm{el}(\mathrm{L}) \simeq B\mathbb{Z}/2$; the reversal acts freely on the nonempty cells of $\square^1$ (no element of a free De Morgan algebra equals its own negation), and L is a $K(\mathbb{Z}/2, 1)$ in both structures.

(Collage) For a sieve–cosieve decomposition of a small category, the nerve is the homotopy colimit of the two parts over the two-sided category of elements of the connecting profunctor (simplicial replacement [1] plus [18]); the formula is natural in functors that preserve the sieve.

Proposition 2.3 (U). *On $\widehat{\square_{\mathrm{DM}}}$ (indeed over any strict test cube category with the cubical-type model structure generated by the maps $m \hat{\otimes} \delta^\varepsilon$), $\mathsf{W}_{\mathrm{type}} \subseteq \mathsf{W}_{\mathrm{test}}$.*

Proof. Since $\square_{\mathrm{DM}}$ is a *strict* test category, the test structure is cartesian: finite products preserve test equivalences [7]. The interval $\square^1$ is representable, so $\mathrm{el}(\square^1)$ has a terminal object and $\square^1$ is aspherical; hence the projection $X \times \square^1 \to X$ lies in $\mathsf{W}_{\mathrm{test}}$ for every X, and by two-out-of-three each endpoint inclusion $X \times \{\varepsilon\} \hookrightarrow X \times \square^1$ is a monomorphism in $\mathsf{W}_{\mathrm{test}}$, i.e. a trivial cofibration of the test structure.

Now let $m : X \hookrightarrow Y$ be any monomorphism and consider the generating map $m \hat{\otimes} \delta^\varepsilon : Y \times \{\varepsilon\} \cup_{X \times \{\varepsilon\}} X \times \square^1 \hookrightarrow Y \times \square^1$. The inclusion of $Y \cong Y \times \{\varepsilon\}$ into its domain is a pushout of the test-trivial cofibration $X \times \{\varepsilon\} \hookrightarrow X \times \square^1$ along $X \times \{\varepsilon\} \hookrightarrow Y \times \{\varepsilon\}$, hence itself a test-trivial cofibration; the composite of this inclusion with $m \hat{\otimes} \delta^\varepsilon$ is the endpoint inclusion $Y \times \{\varepsilon\} \hookrightarrow Y \times \square^1$, again a test-trivial cofibration. By two-out-of-three, $m \hat{\otimes} \delta^\varepsilon \in \mathsf{W}_{\mathrm{test}}$, so every generating trivial cofibration of the type structure is a trivial cofibration of the test structure.

Trivial cofibrations of the test model structure are closed under pushout, transfinite composition, and retract, so every trivial cofibration of the type structure lies in $\mathsf{W}_{\mathrm{test}}$. Trivial fibrations of the type structure have the right lifting property against all monomorphisms, and monomorphisms are exactly the cofibrations of the test structure, so type-trivial fibrations are test-trivial fibrations; in particular they lie in $\mathsf{W}_{\mathrm{test}}$. Finally, an arbitrary $f \in \mathsf{W}_{\mathrm{type}}$ factors as a type-trivial cofibration followed by a type fibration, which by two-out-of-three in the type structure is a type-trivial fibration; both factors lie in $\mathsf{W}_{\mathrm{test}}$, hence so does f. $\square$

Corollary 2.4 (W1). *W is not contractible in the type structure: the map $W \to 1$ is not in $\mathsf{W}_{\mathrm{type}}$.*

Proof. By (U) it would be in $\mathsf{W}_{\mathrm{test}}$, contradicting (W2). $\square$

2.3. **Quotients and strictness.** Throughout, quotients X/G are *strict*: levelwise orbit sets, the colimit of the group action in $\widehat{\square_{\mathrm{DM}}}$. The tension studied in this paper is precisely between the strict quotient and the homotopy quotient: the test structure computes the latter from the former (el turns the strict orbit category into the Borel construction, as in (W2)), while the type structure, as we shall prove, does not.

3. The join model and the equivariant median collage

3.1. **The join and its K-action.** Let $J := \square^1_d \star \square^1_a$ denote the presheaf join: the double mapping cylinder

$$J \;=\; \square^1_d \sqcup_{(\square^1_d\times\square^1_a)\times\{0\}} (\square^1_d\times\square^1_a)\times\square^1_t \sqcup_{(\square^1_d\times\square^1_a)\times\{1\}} \square^1_a,$$

concretely: cells at level m are triples $(\delta,\alpha,\tau)\in \mathrm{DM}(i_1\ldots i_m)^3$, modulo the end collapses ($\tau=0$: the α-component is forgotten; $\tau=1$: the δ-component is forgotten). K acts by

$$\mathrm{sw}\cdot(\delta,\alpha,\tau)=(\delta,\neg\alpha,\tau),\qquad \mathrm{nb}\cdot(\delta,\alpha,\tau)=(\neg\delta,\neg\alpha,\tau),$$

so $g=(\neg\delta,\alpha,\tau)$; the action is by reversals of the two join coordinates, trivial on the cylinder coordinate. The $t{=}0$ end of J/K is $\square^1_d$ modulo the residual reversal, i.e. a copy of L (the *diagonal end* L_D); the $t{=}1$ end is another copy (L_A); on the open cylinder the action is free (no self-negated elements).

3.2. **The median interpolation.**

Definition 3.1. $F(d,a,t) := (d\wedge\neg t)\vee(a\wedge t)\vee(d\wedge a)\in \mathrm{DM}(d,a,t)$.

F interpolates: $F(d,a,0)=d$ and $F(d,a,1)=a$, by absorption. Its two key properties:

Lemma 3.2 (Self-duality)**.** $\neg F(d,a,t)=F(\neg d,\neg a,t)$.

Proof. $\neg F=(\neg d\vee t)\wedge(\neg a\vee\neg t)\wedge(\neg d\vee\neg a)$. Distributing the first two factors gives $(\neg d\wedge\neg a)\vee(\neg d\wedge\neg t)\vee(t\wedge\neg a)\vee(t\wedge\neg t)$; meeting with $(\neg d\vee\neg a)$ and absorbing (each of the last three joinands is below $\neg d$ or $\neg a$ up to the terms retained) yields $(\neg d\wedge\neg t)\vee(\neg a\wedge t)\vee(\neg d\wedge\neg a)=F(\neg d,\neg a,t)$. ◻

Lemma 3.3 (Flow identity)**.** *$F(d,a,t\wedge s)=F(d,F(d,a,t),s)$ and dually $F(d,a,t\vee s)=F(F(d,a,t),a,s)$.*

Proof. Expand the right side with $F:=F(d,a,t)$: $(d\wedge\neg s)\vee(F\wedge s)\vee(d\wedge F)=(d\wedge\neg s)\vee(d\wedge\neg t\wedge s)\vee(a\wedge t\wedge s)\vee(d\wedge a\wedge s)\vee(d\wedge\neg t)\vee(d\wedge a\wedge t)\vee(d\wedge a)$. Absorption removes the three joinands below $d\wedge\neg t$ or $d\wedge a$, leaving $(d\wedge(\neg s\vee\neg t))\vee(a\wedge t\wedge s)\vee(d\wedge a)=F(d,a,t\wedge s)$. The second identity is dual (Lemma 3.2). ◻

3.3. **The collage map.**

Definition 3.4. $\Phi := (F(d,a,t),\ F(d,\neg a,t)) : \square^3\to\square^2$.

Proposition 3.5. *Φ intertwines the K-actions: $\mathrm{sw}\circ\Phi=\Phi\circ\mathrm{sw}_J$ and $\mathrm{nb}\circ\Phi=\Phi\circ\mathrm{nb}_J$, and Φ respects the end collapses. Hence Φ descends to*

$$\bar\Phi : J/K \longrightarrow W.$$

Moreover $\bar\Phi$ restricts to isomorphisms of the ends onto the stabilized layers: the $t{=}0$ end L_D maps isomorphically onto the diagonal layer ($\delta\mapsto(\delta,\delta)$), and the $t{=}1$ end onto the antidiagonal layer ($\alpha\mapsto(\alpha,\neg\alpha)$).

Proof. sw-equivariance is immediate from the definition (the two components swap when $a \mapsto \neg a$). nb-equivariance is Lemma 3.2 applied in each component. At $t = 0$ both components equal d (absorption), so the α-collapse is respected and the end maps to the diagonal; at $t = 1$ the components are $(a, \neg a)$. The residual reversals match on both sides, giving isomorphisms of the end layers. □

Geometrically, $\bar{\Phi}$ reads a point of the Klein-quotient square in *join coordinates*: distance from the diagonal is the cylinder coordinate, and the median convex combination interpolates between the two fixed lines. The map is a presheaf-level realization of the collage that computes $\mathrm{el}(W)$ in (W2).

4. The test side: $\mathrm{el}(\bar{\Phi})$ is a weak equivalence

Theorem 4.1. $\mathrm{el}(\bar{\Phi}) : \mathrm{el}(J/K) \to \mathrm{el}(W)$ *is a weak homotopy equivalence. Hence* $\bar{\Phi} \in \mathsf{W}_{\mathrm{test}}$.

The proof occupies this section. Write $\phi := \mathrm{el}(\bar{\Phi})$. Recall from (W2) the sieve–cosieve decomposition of $\mathrm{el}(W)$: the sieve $S_W =$ the stabilized layers (diagonal $\sqcup$ antidiagonal, with no morphisms between them), the cosieve $U_W =$ the free layer.

4.1. The pulled-back decomposition.

Since the preimage of a sieve under a functor is a sieve, set $S'' := \phi^{-1}(S_W)$, $U'' := \phi^{-1}(U_W)$. Sieves of a category of elements correspond to subpresheaves; S'' corresponds to the *collapse locus* $Z = Z_D \sqcup Z_A \subseteq J/K$:

$$Z_D := \{(\delta, \alpha, \tau) : F(\delta, \alpha, \tau) = F(\delta, \neg\alpha, \tau)\}, \quad Z_A := \{\cdots = \neg F(\delta, \neg\alpha, \tau)\},$$

equational, hence substitution-closed, conditions. (Z_D contains the t=0 end, where both components equal δ; Z_A contains the t=1 end.) By naturality of the collage formula (Collage) in sieve-preserving functors, it suffices to prove that the three legs of

$$\begin{array}{ccccc} NS'' & \longleftarrow & N\hat{W}'' & \longrightarrow & NU'' \\ \downarrow & & \downarrow & & \downarrow \\ NS_W & \longleftarrow & N\hat{W} & \longrightarrow & NU_W \end{array}$$

are weak equivalences.

4.2. Leg one: the collapse locus retracts onto the ends.

Lemma 4.2. Z_D *is closed under lowering* τ *(if* $(\delta, \alpha, \tau) \in Z_D$ *then* $(\delta, \alpha, \tau \wedge s) \in Z_D$ *for every* s*), and dually* Z_A *under raising* τ*. Moreover* $Z_D \cap \{t{=}1\} = \emptyset = Z_D \cap Z_A$.

Proof. By the flow identity, $F(\delta, \alpha, \tau \wedge s) = F(\delta, F(\delta, \alpha, \tau), s)$, and likewise with $\neg\alpha$; if the two inner values agree, so do the outer ones. At $t = 1$ the components are α and $\neg\alpha$, never equal (no self-negated elements); membership in both loci would force $F = \neg F$, likewise impossible. □

Consequently the presheaf cylinder $h_s : (\delta, \alpha, \tau) \mapsto (\delta, \alpha, \tau \wedge \neg s)$ restricts to Z_D and deformation-retracts it onto the t=0 end, fixing the end pointwise; dually for Z_A. A presheaf homotopy induces, after $N\mathrm{el}(-)$, equality of the maps in the homotopy category (the projection $X \times \square^1 \to X$ is in $\mathsf{W}_{\mathrm{test}}$). Combining with Proposition 3.5 (the end maps are isomorphisms of layers): $\phi|_{Z_D} \simeq$ (iso) $\circ$ (retraction), a weak equivalence; the A-component is dual; the two components are disjoint as categories on both sides.

4.3. **Legs two and three: middle layers via terminal covers.**

Lemma 4.3 (Free quotients of contractible covers)**.** *Let G be a finite group acting on $X \in \widehat{\square_{\mathrm{DM}}}$ such that the action on the nonempty cells is free, and suppose the full subcategory $\mathcal{C} \subseteq \mathrm{el}(X)$ on a G-stable class of cells has contractible nerve. Then: (i) $\mathrm{el}(X/G) \cong \mathrm{el}(X)/G$ as categories, and the corresponding full subcategory of $\mathrm{el}(X/G)$ is the quotient category $\mathcal{C}/G$; (ii) the action of G on $\mathcal{C}$ is free on objects and morphisms, $N(\mathcal{C}/G) \cong N(\mathcal{C})/G$, and the quotient map $N\mathcal{C} \to N(\mathcal{C}/G)$ is a principal G-covering of simplicial sets; hence $N(\mathcal{C}/G) \simeq BG$; (iii) if $\mathcal{C} \to \mathcal{D}$ is a G-equivariant functor between two such subcategories (over possibly different G-presheaves), the induced functor $\mathcal{C}/G \to \mathcal{D}/G$ is a weak equivalence.*

Proof. (i) Cells of X/G are orbits of cells and a morphism $([m], [x]) \to ([n], [x'])$ is a cube map f with $[x] = [x'] \cdot f$, i.e. $x \cdot g = x' \cdot f$ for a unique g (freeness); this is exactly a morphism of the quotient category. (ii) Freeness on objects gives freeness on morphisms (a morphism fixed by g has fixed source); a free action of a finite group on a small category quotients to a category whose nerve is the quotient simplicial set, and a free simplicial action has quotient a principal covering; since $N\mathcal{C}$ is contractible, the quotient is a $K(G, 1)$. (iii) The equivariant functor induces a map of principal G-coverings over the quotients, hence a π_1-isomorphism ($= \mathrm{id}_G$ after the deck identifications) between $K(G, 1)$'s, and both quotients are aspherical by (ii); apply Whitehead. ☐

Lemma 4.4. *ϕ restricts to weak equivalences $U'' \to U_W$ and $\hat{W}'' \to \hat{W}$.*

Proof. U'' is the full subcategory of cells not in Z. Upstairs (before quotienting) the corresponding full subcategory of $\mathrm{el}(\square^3$-cylinder) contains the generic cell $\mathrm{id}_{[3]}$ (its Φ-components are separated: at (d, a, t) generic the antichain normal forms differ), and every object maps to it uniquely, so the cover is contractible; K acts freely on the relevant cells (no self-negated elements). The same holds for the free-layer cover of W with terminal object $\mathrm{id}_{[2]}$. Lemma 4.3 now applies verbatim: both quotients are BK's and the equivariant comparison is a weak equivalence. For $\hat{W}''$ we use the same comma-piece analysis as in Appendix A. By construction the two-sided category of the connecting profunctor is the arrow category: objects are triples (a sieve cell s, a cosieve cell u, a map $s \to u$), and morphisms are commuting squares. Lifting along the free K-covers of the two parts identifies it with the quotient by K of the corresponding arrow category $\mathcal{A}$ upstairs, on which K acts freely through the cosieve component. The projection of $\mathcal{A}$ onto its sieve component is a Grothendieck fibration whose fiber over s is the coslice $(s \downarrow F'')$ into the cosieve cover; each such coslice has a terminal object, namely the generic cell with its canonical map $(\mathrm{id}, s \to \mathrm{id})$, hence is contractible. The base — the sieve part upstairs — contracts by the flow retraction of Lemma 4.2 onto the end, which is the category of elements of a representable and hence contractible. By Quillen's Theorem A, $N\mathcal{A}$ is contractible, so $N\hat{W}'' \cong N\mathcal{A}/K \simeq BK$ as in Lemma 4.3(ii). The same analysis applies to $\hat{W}$ on the target side, and the comparison induced by ϕ is an equivariant map of contractible free covers, hence a weak equivalence by Lemma 4.3(iii). □

This completes the proof of Theorem 4.1. (The statement is also consistent with a direct computation: the collage of $\mathrm{el}(J/K)$ evaluates to $\mathrm{hocolim}(B\mathbb{Z}/2 \leftarrow BK \to B\mathbb{Z}/2) = B\mathbb{Z}/2 \star B\mathbb{Z}/2$, matching (W2).)

5. Strictification and the parity lemmas

5.1. Maps out of a strict quotient.

Lemma 5.1 (Strictification)**.** *For every $Y \in \widehat{\square_{\mathrm{DM}}}$, precomposition with the quotient of the Yoneda embedding induces a bijection*

$$\widehat{\square_{\mathrm{DM}}}(W, Y) \;\cong\; \{\, s \in Y([2]) \;:\; s \cdot \mathrm{sw} = s,\; s \cdot \mathrm{nb} = s \,\},$$

where $k \in K \subseteq \square_{\mathrm{DM}}([2],[2])$ acts by restriction. Likewise $\widehat{\square_{\mathrm{DM}}}(J/K, Y)$ is the set of elements of $Y([3])$ invariant under K_J and compatible with the end collapses, and $\widehat{\square_{\mathrm{DM}}}(W \times \square^1, Y)$ is the set of K-invariant elements of $Y([2{+}1])$ (invariance in the first two coordinates).

Proof. W is the coequalizer of the K-action on $\square^2$; apply Yoneda. The same for the other two (for the cylinder, $(\square^2 \times \square^1)/(K \times 1) = W \times \square^1$ since quotients are levelwise). □

Thus homotopy-theoretic questions about maps from W become questions about strictly invariant cells — finitely constrained data.

5.2. The parity lemmas.

The next two lemmas are the heart of the paper. Recall the representation (1): elements of DM_2 are monotone functions on the literal cube $\{0,1\}^4$ with coordinates $(v_x, v_{\neg x}, v_y, v_{\neg y})$; negation is $(\neg\varphi)(p) = 1 - \varphi(\rho p)$; the generator swap sw acts by exchanging the pairs; and $\mathrm{lswap} \circ \rho = c$ is the bitwise complement, so that

$$\varphi^{\mathrm{nb}} = \neg\varphi \iff \varphi(c\,q) = 1 - \varphi(q) \quad \forall q. \tag{2}$$

Lemma 5.2 (First parity lemma)**.** *No $\varphi \in \mathrm{DM}_2$ satisfies $\varphi^{\mathrm{sw}} = \neg\varphi$.*

Proof. The condition reads $\varphi(\mathrm{sw}\,p) = 1 - \varphi(\rho\,p)$ for all p. The point $q = (1,0,1,0)$ satisfies $\mathrm{sw}\,q = q$ and $\rho\,q = q$; at q the condition becomes $\varphi(q) = 1 - \varphi(q)$. □

Lemma 5.3 (Second parity lemma)**.** *No $\varphi \in \mathrm{DM}_2$ satisfies both $\varphi^{\mathrm{sw}} = \varphi$ and $\varphi^{\mathrm{nb}} = \neg\varphi$.*

Proof. Let $m = (1,0,0,1)$. Its bitwise complement is $c\,m = (0,1,1,0) = \mathrm{sw}\,m$. Then symmetry gives $\varphi(c\,m) = \varphi(m)$ while (2) gives $\varphi(c\,m) = 1 - \varphi(m)$. □

5.3. **Classification of invariant cells.** Cells of $(J/K)([2])$ are, per the end normalization: end cells ($\tau \in \{0,1\}$, one component retained modulo the residual reversal) and cylinder classes $[(\delta,\alpha,\tau)]$, the K_J-orbit $\{(\pm\delta,\pm\alpha,\tau)\}$ (all four sign patterns).

Proposition 5.4 (Rigidity of invariant cells)**.** *Let $[(\delta,\alpha,\tau)]$ be a cylinder class of $(J/K)([2])$ invariant under both coordinate substitutions* $\mathrm{sw}, \mathrm{nb} \in \square_{\mathrm{DM}}([2],[2])$*. Then δ, α, τ are separately strictly invariant: $\varphi^{\mathrm{sw}} = \varphi = \varphi^{\mathrm{nb}}$ for each component φ.*

Proof. Invariance under sw means $(\delta^{\mathrm{sw}}, \alpha^{\mathrm{sw}}, \tau^{\mathrm{sw}}) = (\pm\delta, \pm\alpha, \tau)$ for one of the four sign patterns. A minus sign on either component is excluded by Lemma 5.2; hence $\delta^{\mathrm{sw}} = \delta$, $\alpha^{\mathrm{sw}} = \alpha$, $\tau^{\mathrm{sw}} = \tau$. Invariance under nb now reads $(\delta^{\mathrm{nb}}, \alpha^{\mathrm{nb}}, \tau^{\mathrm{nb}}) = (\pm\delta, \pm\alpha, \tau)$ with the components already sw-symmetric; a minus sign is excluded by Lemma 5.3. □

Remark 5.5. Proposition 5.4 was found, and is independently confirmed, by an exhaustive machine census: of the 168^3 triples ($|\mathrm{DM}_2| = 168$, the Dedekind number $M(4)$), the census finds 372 invariant classes in total, of which 360 are interior cylinder classes — every one admitting a strictly invariant representative — and the remaining 12 lie on the collapsed ends (Section 13). The same census shows, as context, that no strict section-like phenomena intervene: no invariant class maps under $\bar{\Phi}$ to the generic cell of W.

6. The nullity theorem

Fix the algebraic fibrant replacement $u : J/K \to R$ of Lemma 2.2. Since every object is cofibrant and R is fibrant, homotopy classes $[W, R]$ are strict maps modulo cylinder homotopy, and by Lemma 5.1 both maps and homotopies are strictly invariant cells.

The inductive engine is the following observation. Write $\sim$ for the equivalence relation on index pairs (B, c) generated by the boundary identifications (i) and the naturality reindexings (ii) of Lemma 2.2, so that two filler values coincide exactly when their index data are related; and note that the presheaf action on filler values is $\mathrm{val}(B,c) \cdot f = \mathrm{val}(B, c \cdot f)$. For a box B of shape $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$ and a cell $c = (c_1, \tau)$ of $\square^n \times \square^1$ define the *slide*

$$c_z \;:=\; (c_1,\ \tau \wedge \neg z) \quad (\varepsilon = 0), \qquad c_z \;:=\; (c_1,\ \tau \vee z) \quad (\varepsilon = 1),$$

a cell of $\square^n \times \square^1$ in one further variable z.

Lemma 6.1 (Equivariant filler slide)**.** *The slide is compatible with the relation: $(B, c) \sim (B', c')$ implies $(B, c_z) \sim (B', c'_z)$. Consequently, if* $\mathrm{val}(B, c)$ *is invariant under a group of substitutions k acting on its level (i.e. $(B, c \cdot k) \sim (B, c)$ for all k), then so is* $\mathrm{val}(B, c_z)$ *for every z ($(c \cdot k)_z = c_z \cdot k$, as k does not involve z), and the slide is a homotopy from* $\mathrm{val}(B, c)$ *at $z = 0$ to a cell of the box itself — earlier-stage material — at $z = 1$.*

Proof. It suffices to check the generating relations. (i) *Boundary*: the box domain is $S \times \square^1 \cup \square^n \times \{\varepsilon\}$. If c_1 lies in S then so does the slid cell's first component (it is unchanged); if $\tau = \varepsilon$ then $\tau \wedge \neg z = 0 = \varepsilon$ (resp. $\tau \vee z = 1 = \varepsilon$): the slide preserves the box domain, on which values are the prescribed old cells, and there the relation is equality of old cells, preserved because the slide acts by a substitution. (ii) *Naturality*: the reindexings have the form $(B \cdot \hat{\sigma},\, c') \sim (B,\, \hat{\sigma} \circ c')$ with $\hat{\sigma} = \sigma \times \mathrm{id}_{\square^1}$ a box morphism over a substitution σ of the base shape; since $\hat{\sigma}$ does not touch the cylinder coordinate, $\hat{\sigma} \circ (c'_z) = (\hat{\sigma} \circ c')_z$, and the relation slides. Chains of generators slide term by term. For the last clause: at $z = 1$ the slid cell is $(c_1, 0)$ (resp. $(c_1, 1)$), which lies in $\square^n \times \{\varepsilon\}$, i.e. in the box domain, so the value is a cell prescribed by B over the earlier stage. $\square$

Remark 6.2. Lemma 6.1 deliberately avoids analysing *how* the invariance of a filler value arises. It may arise with $c \cdot k = c$; but it may equally arise through the naturality relation — for instance when the box itself is invariant under an automorphism of its base shape intertwining k — and no coordinatewise rigidity can be inferred. The slide transports the entire relation chain, whatever it is, down the filling cylinder.

Theorem 6.3 (Nullity)**.** *Every map $f : W \to R$ is homotopic to a constant map.*

Proof. By Lemma 5.1, f corresponds to a K-invariant element $s \in R([2])$. The cells of R are filtered by the stages of the construction of Lemma 2.2; equality of cells is detected at a stage. We induct on the stage at which s appears.

Base of the induction: s is a cell of J/K. Three cases.

- s is a cylinder class. By Proposition 5.4 it has strictly invariant components, so the classifying map $(\delta, \alpha, \tau) : \square^2 \to \square^3$ is K-invariant on the nose and descends to $\bar{\sigma} : W \to \square^3$ with f factoring through the representable $\square^3$; the latter contracts to a vertex by the connection homotopy $(u_1, u_2, u_3, z) \mapsto (u_1 \wedge \neg z, u_2 \wedge \neg z, u_3 \wedge \neg z)$, a strict presheaf cylinder, so f is homotopic to a constant.
- s is an end cell, say in L_D. Then f factors through the end inclusion $\mathrm{L}_D \hookrightarrow J/K$, which is null-homotopic by the explicit slide $H(\varphi, \zeta) := [(\varphi, 0, \zeta)]$, a presheaf map $\mathrm{L}_D \times \square^1 \to J/K$ with H_0 the inclusion and H_1 constant. Dually for L_A.
- s is a vertex class: f is constant.

Inductive step: s is a fresh filler value. Write $s = \mathrm{val}(B, c)$ as in Lemma 2.2, with $c = (c_1, \tau) \in (\square^n \times \square^1)([2])$. Invariance means $(B, c \cdot k) \sim (B, c)$ for all $k \in K$. Let $H_z := \mathrm{val}(B, c_z)$ be the slide of Lemma 6.1, formed at level 2+1 (the slide of the level-3 cell $(c_1 \circ \mathrm{pr}, \tau \wedge \neg z)$). By Lemma 6.1 the relation chains witnessing invariance transport to every z, so H is a K-invariant element of $R([2{+}1])$, i.e. by Lemma 5.1 a homotopy of maps $W \to R$; it interpolates from $H_0 = s$ to H_1, a cell prescribed by the box over an earlier stage, and H_1 is itself K-invariant (restrict the invariant H at $z = 1$). The induction hypothesis applied to H_1 finishes the step.

Limit stages add no cells. This exhausts all cases. $\square$

Remark 6.4. The proof uses no homology and no analysis of the homotopy type of R; and it is uniformity that carries it. On a Reedy-type site one could try to run a similar induction with a naive fibrant replacement, but on $\square_{\mathrm{DM}}$ the cell shapes of monomorphisms include quotients of representables — including Klein-quotient shapes, i.e. copies of the very object W — and an induction over such material would be circular. Uniform filling confines the fresh material to representable-shaped filler values (Lemmas 2.1 and 2.2), where the parity lemmas leave strict invariance no room: every invariant cell slides down its own filling cylinder onto earlier material, and at the bottom the strict quotient's cells factor through explicitly contractible or explicitly null pieces. The contrast with the test side, where the category of elements absorbs the quotient into a Borel construction, is the entire content of the separation.

7. The separation

Theorem 7.1. $\bar{\Phi} \in \mathsf{W}_{\mathrm{test}} \setminus \mathsf{W}_{\mathrm{type}}$. *Consequently* $\mathsf{W}_{\mathrm{type}} \subsetneq \mathsf{W}_{\mathrm{test}}$.

Proof. $\bar{\Phi} \in \mathsf{W}_{\mathrm{test}}$ is Theorem 4.1. Suppose $\bar{\Phi} \in \mathsf{W}_{\mathrm{type}}$. Let $u : J/K \to R$ and $u' : W \to R'$ be fibrant replacements. Then $R(\bar{\Phi}) : R \to R'$ is a weak equivalence between bifibrant objects, hence a homotopy equivalence; let $h : R' \to R$ be a homotopy inverse. The composite $g := h \circ u' : W \to R$ satisfies $R(\bar{\Phi}) \circ g \simeq u'$. By Theorem 6.3, g is homotopic to a constant, so u' is homotopic to a constant map. Since u' is a weak equivalence, this makes $\mathrm{id}_{R'}$ factor through a point in the homotopy category, i.e. R' — and hence W — is contractible in the type structure, contradicting Corollary 2.4. $\square$

Corollary 7.2. *In the type structure, W admits no essential map to the type realization of $B\mathbb{Z}/2 \star B\mathbb{Z}/2$: writing R for a fibrant replacement of J/K (whose test type is the join), $[W, R] = \{*\}$. In particular the type homotopy type of W is neither contractible nor equivalent to the join: it is a De Morgan cubical set on which the two structures genuinely disagree.*

Corollary 7.3 (The isotropy landscape)**.** *Combining with* [15]*: for isotropy quotients of De Morgan cubes, the free and the fixed-point-full families are agreement families (both classifying spaces, respectively both contractible, in the two structures), whereas the mixed family is* non-uniform*: it contains* *the first separating examples, beginning with the Klein quotient of the square, as well as further agreement cases. The complete classification of the mixed family in dimensions up to three — 78 agreement cases and 20 separations among the 98 subgroups of the 3-cube — is carried out in the companion paper [16], which builds on the present results.*

Remark 7.4 (What the type structure sees)**.** The corollaries delimit, but do not compute, the type homotopy type of W. The strictification lemma reduces its homotopy and cohomology to computations with invariant cells of fibrant replacements of *other* objects, which the parity mechanism again constrains severely; a full determination of type(W) — an explicit genuinely non-test homotopy type isolated in this setting — is an interesting open problem. We note only that all four candidate "twist" squares constructed from the loops of the two ends are null (they factor through one-dimensional segment material), so the type structure truncates the mixed quotient far more brutally than the test structure.

8. The fundamental group of W sees no difference

This section records a negative result which motivates the construction and calibrates the machinery.

The cells of $W([1])$ — pairs (φ, ψ) over $\mathrm{DM}(()u)$ modulo K — comprise twelve classes, of which eight are invariant under the reversal $u \mapsto \neg u$ and hence are strict maps from the reversal circle L to W. Among them are the images of the two strata circles: the diagonal loop $\delta = [(u, u)]$ at the vertex $v_0 = [(0, 0)]$ and the antidiagonal loop $\gamma = [(u, \neg u)]$ at $v_1 = [(0, 1)]$.

Proposition 8.1. *No cell of $W([2])$ has boundary δ on one face and constants on the other three (in any orientation): the loop δ bounds no strict disc. (*168^2 *states, machine search.)*

Proposition 8.2. *Nevertheless δ is freely null-homotopic through reversal-invariant squares: the single invariant cell $\big(u \wedge v,\ u \vee \neg v\big)$ has $v{=}1$-face δ and $v{=}0$-face the constant edge at the* other *vertex v_1. The full free-homotopy graph on the eight invariant edge classes, with one edge for each invariant square, is connected. The same holds for γ.*

Thus $[L, R(W)]$ carries no obstruction from the strata loops: the based rigidity of Proposition 8.1 is dissolved by a free slide across the quotient. Any object-level difference between the two structures on this site must therefore live above the fundamental group, and any proof must block such slides structurally. The subdivided cone of Section 10 is designed exactly so that the corresponding slides (the cone slides) can only *help*: in C they null-homotope the cone sector and normalize the problem, rather than dissolving it.

9. Level-three cells as a satisfiability problem

Cells of W at level three are pairs over $\mathrm{DM}(()u, v, w)$, a free De Morgan algebra with 7,828,354 elements; exhaustive search is impossible. We record a method that decides existence questions about such cells exactly.

Proposition 9.1. *Let a boundary condition (prescribed faces, up to the K-orbit) and an invariance condition (class-invariance under prescribed substitutions, with deck twists) be given for a pair (A, B) over $\mathrm{DM}(()u, v, w)$. Existence of (A, B) is decidable by a finite search: branching over the 4^4 choices of face representatives and deck twists, each branch reduces to (i) a union-find with parity over the 128 value slots $A(p), B(p)$, $p \in L_3 = \{0,1\}^6$, where the De Morgan negation acts by $(\neg A)(p) = 1 - A(\rho p)$; (ii) fixed values on the face planes; and (iii) monotonicity, a system of implications over the 665 comparable pairs of L_3, i.e. a 2-SAT instance.*

The point is that a monotone map is an arbitrary monotone $\{0,1\}$-labelling of the literal cube, so all conditions in sight are pointwise-linear or order-implications; no normal-form or enumeration of $\mathrm{DM}(()3)$ is needed. Two computations made with this method are used later:

Proposition 9.2. *W has exactly 72 strict endomorphism classes (K-invariant elements of $W([2])$). The strict-homotopy component of the identity consists of exactly 3 classes — the identity and two median deformations — and contains no constant. The 42 classes that factor strictly through $\bar{\Phi}$ (through the interior of J/K or through its two ends) form a family disjoint from the identity's component.*

That the identity's component avoids the constants is forced by Corollary 2.4 and serves as a calibration of the method; that it avoids the $\bar{\Phi}$-factoring family is the stage-zero shadow of Theorem 7.1.

10. The subdivided cone

Construction 10.1. Let $\widetilde{C}$ be the presheaf obtained by gluing, in order: the square $\square^2$; the mapping cylinder $J \times \square^1$ attached along $t \mapsto \Phi$ at $\tau_1 = 0$; a second cylinder $J \times \square^1$ attached to the first along the middle copy J (the $\tau_1 = 1$ end identified with the $\tau_2 = 0$ end); and the collapse of the far end $\tau_2 = 1$ to a vertex pt. The group K acts compatibly (standardly on $\square^2$, through the twisted action on both J-cylinders, fixing pt), and

$$C := \widetilde{C}/K.$$

C is a finite presheaf; we call the images of the three regions the *base sector* (W), the *cylinder sector*, and the *cone sector*.

The subdivision has one purpose, recorded now:

Lemma 10.2 (Sector separation). *Every cell of $\widetilde{C}$ is supported in $\widetilde{M} := \square^2 \cup_\Phi (J \times \square^1)$ (no cone-sector support) or in $\mathrm{cone}(J) := (J \times \square^1) \cup \mathsf{pt}$ (no* *base-sector support); the interface of the two sieves is the middle copy of J. On $\widetilde{M}$ the strict retraction $\tilde{r}(j,\tau_1) = \Phi(j)$, $\tilde{r}|_{\square^2} = \mathrm{id}$, is totally defined and K-equivariant; on* $\mathrm{cone}(J)$ *the strict cone slide $(j,\tau_2) \mapsto (j, \tau_2 \vee s)$ is totally defined and equivariant, and contracts the sector to the vertex.*

Proof. A cell of the first cylinder has τ_1-faces in $\square^2$ (at 0) and the middle J (at 1); a cell of the second has τ_2-faces in the middle J and the vertex. No single cell carries both a $\square^2$-face and a vertex face, which is the only claim with content; the displayed maps are the evident ones, and equivariance is inherited from the equivariance of Φ and of the cylinder coordinates. $\square$

Proposition 10.3. *C is test-contractible.*

Proof. All attachments in Construction 10.1 are along monomorphisms — the end inclusions $J/K \times \{\varepsilon\} \hookrightarrow J/K \times \square^1$ — and monomorphisms are the cofibrations of both structures; note that $\bar{\Phi}$ itself is *not* assumed a cofibration, and never used as one. The first cylinder yields the mapping cylinder M of $\bar{\Phi}$, which strictly deformation retracts onto W; the remaining part is the cone of J/K, which strictly deformation retracts onto its vertex; both retractions are weak equivalences in any model structure. Thus C is the pushout of the diagram $M \hookleftarrow J/K \hookrightarrow \mathrm{cone}(J/K)$ with both legs cofibrations, hence a homotopy pushout by left properness, and

$$C \;\simeq_{\text{test}}\; \mathrm{hocolim}\big(W \xleftarrow{\;\bar{\Phi}\;} J/K \longrightarrow *\big) \;\simeq_{\text{test}}\; \mathrm{hocolim}\big(W \xleftarrow{\;\sim\;} W \to *\big) \;\simeq\; *,$$

using $\bar{\Phi} \in \mathsf{W}_{\text{test}}$ (Theorem 4.1) and the invariance of homotopy pushouts under weak equivalences of diagrams. (The subdivision inserts one more cylinder, treated identically.) $\square$

11. Strictification, sectors, and projections

By the strictification lemma (Lemma 5.1), maps $W \to R(C)$ correspond to K-invariant elements of $R(C)([2])$, and homotopies between them to invariant elements one level up. The analysis of such data is organized by the K-cover $\widetilde{C}$ of C, through a dictionary that we now record once and for all. For a finite group G acting on a presheaf $\widetilde{X}$ we write $X := \widetilde{X}/G$ (levelwise orbit sets) and

$$R_G(\widetilde{X}) \;:=\; R(\widetilde{X}/G),$$

the algebraic replacement (Lemma 2.2) of the quotient; we call it the *G-equivariant replacement of $\widetilde{X}$*, a name justified by the following lemma.

Lemma 11.1 (Equivariant descent)**.** *Let a finite group G act on $\widetilde{X}$, and let $\widetilde{Z} \subseteq \widetilde{X}$ be a G-stable subpresheaf with image $Z := \widetilde{Z}/G \subseteq X$.*

(1) Cells. *Stage-zero cells of $R_G(\widetilde{X})$ at level $[n]$ are the G-orbits of cells of $\widetilde{X}([n])$; every fresh cell is a filler value* $\mathrm{val}(b,c)$ *of a box datum b over an earlier stage, presented by representable-shaped data via the pullback in the proof of Lemma 2.1. Call b* supported in Z *if all its prescribed cells lie in the (inductively defined) Z-part of its stage.*

(2) Subreplacements. *The canonical map* $R(Z) \to R(X)$ *is injective, and its image — the* $\widetilde{Z}$-sector *— consists exactly of the old cells of* Z *and the filler values of* Z*-supported boxes. Identifications among sector cells computed in* $R(X)$ *coincide with those of* $R(Z)$*.*

(3) Projections. *A strict* G*-equivariant map* $\tilde{f} : \widetilde{X} \to \widetilde{Y}$ *descends to* $f : X \to \widetilde{Y}/G$ *and induces* $R(f) : R_G(\widetilde{X}) \to R_G(\widetilde{Y})$*, which carries the* $\widetilde{Z}$*-sector into the* $\tilde{f}(\widetilde{Z})$*-sector. Moreover, for any subgroup* $H \leq \square_{\mathrm{DM}}([n],[n])^{\times}$ *of invertible substitutions,* $R(f)$ *sends* H*-invariant* n*-cells to* H*-invariant* n*-cells, and likewise for cylinder homotopies; here invariance is under the* source-coordinate *substitution action, the notion produced by Lemma 5.1, and is not to be confused with the covering group* G*, which acts on* $\widetilde{X}$ *but has no residual action on the quotient replacement. In particular, for* $W = \square^2/K$ *(where* $H = K$ *acting on* $[2]$*), post-composition with* $R(f)$ *carries* $[W, R(X)]$*-data to* $[W, R(\widetilde{Y}/K)]$*-data.*

Proof. (1) is the presentation of Lemma 2.2 applied to X, combined with the pullback of boxes to representable shapes in the proof of Lemma 2.1; quotients are levelwise, so stage zero is the orbit presheaf.

(2) Functoriality of the algebraic construction along the inclusion $Z \hookrightarrow X$ sends stages to stages and box data to box data, giving the canonical map, with the stated image. The substance is injectivity: two cells of $R(Z)$ identified in $R(X)$ must already be identified in $R(Z)$. By Lemma 2.2(2), identifications in $R(X)$ are generated by the relations (i) and (ii). A generator of type (i) relates a filler value of a box b to a prescribed cell of the same b; when b is Z-supported this is a relation of $R(Z)$. A generator of type (ii) can, however, relate two sector cells through a box b that is *not* Z-supported:

$$\mathrm{val}(b \circ \hat{\sigma}_1, c_1) \;=\; \mathrm{val}(b, \hat{\sigma}_1 c_1) \;=\; \mathrm{val}(b, \hat{\sigma}_2 c_2) \;=\; \mathrm{val}(b \circ \hat{\sigma}_2, c_2)$$

with $\hat{\sigma}_1 c_1 = \hat{\sigma}_2 c_2$ and both restricted boxes Z-supported. Such a step normalizes through the common restriction. Write the common cell as a pair of substitutions, $\hat{\sigma}_i c_i = \langle d_0, d_1 \rangle$ (the site has diagonals); then $c_i = \langle c_{i0}, d_1 \rangle$ with $\sigma_i \circ c_{i0} = d_0$, and relation (ii) applied *inside* each restricted box gives

$$\mathrm{val}(b \circ \hat{\sigma}_i,\, c_i) \;=\; \mathrm{val}\big((b \circ \hat{\sigma}_i) \circ \hat{c}_{i0},\, \langle \mathrm{id}, d_1 \rangle\big),$$

$$(b \circ \hat{\sigma}_1) \circ \hat{c}_{10} \;=\; b \circ \widehat{d_0} \;=\; (b \circ \hat{\sigma}_2) \circ \hat{c}_{20}$$

(with pulled-back box subpresheaves throughout), and $b \circ \widehat{d_0}$, being a restriction of the Z-supported box $b \circ \hat{\sigma}_1$, is itself Z-supported. An arbitrary chain of generating relations between two sector cells is reduced by successively eliminating its maximal excursions through non-Z-supported boxes. Along an excursion, relations of type (ii) preserve the absolute position in the codomain of the ambient box, so an excursion that enters or leaves through a relation of type (i) has old cells at both ends sitting at the same position, and these are equal by the evaluation compatibility of restricted boxes (Lemma 2.2(2)) — such an excursion is deleted outright. An excursion that enters and leaves through relations of type (ii) is replaced by the common-restriction pair displayed above. Each elimination strictly decreases the number of non-Z-supported boxes visited, so the process terminates. Thus every identification between sector cells is generated by relations among Z-supported boxes, which are precisely the relations of $R(Z)$.

(3) Functoriality along the strict map f: $R(f)$ carries a Z-supported box to an $\tilde{f}(\widetilde{Z})$-supported box. For the invariance statement, $R(f)$ is a map of presheaves, hence commutes with the right action of every substitution $h \in H$ on cells; if $x \cdot h = x$ then $R(f)(x) \cdot h = R(f)(x \cdot h) = R(f)(x)$, and the same computation applies one level up for cylinder homotopies. The last statement is Lemma 5.1. $\square$

Write ι for the tautological class (the generating cell of $\square^2$), and let

$$\widetilde{M} := \square^2 \cup_\Phi (J \times \square^1), \qquad \mathrm{cone}(J) := (J \times \square^1) \cup \mathsf{pt}$$

be the two closed sectors of Lemma 10.2, with interface the middle copy of J.

Lemma 11.2 (Sector replacements and projections). (1) *The cells of the equivariant replacement whose boxes are supported in $\widetilde{M}$ form a subreplacement canonically isomorphic to $R_K(\widetilde{M})$, and likewise for* cone(J)*; call these the $\widetilde{M}$-*sector *and the* cone sector.

(2) *The strict equivariant retraction $\tilde{r}\colon \widetilde{M} \to \square^2$ induces $R_K(\widetilde{M}) \to R_K(\square^2)$; it sends invariant classes to invariant classes and invariant homotopies to invariant homotopies, and fixes the base pointwise. Under descent, invariant $\widetilde{M}$-sector data projects to $[W, R(W)]$-data.*

(3) *The strict equivariant projection $\pi\colon J \times \square^1 \to J$ of the second cylinder induces, on cone-sector cells with no vertex support, a projection to $R_K(J)$-data; under descent, invariant interface data projects to $[W, R(J/K)]$-data.*

(4) *Every cell of the second cylinder with vertex support has all faces in the cone sector; no cell has both base and vertex support (Lemma 10.2).*

Proof. (1) is Lemma 11.1(2) applied to the two K-stable closed sieves of Lemma 10.2. (2) and (3) are Lemma 11.1(3) applied to the strict equivariant maps $\tilde{r}$ and π. (4) is Lemma 10.2. $\square$

We record for later use, as a secondary observation, that invariant classes carry stagewise *twists* $\delta \in \mathrm{Hom}(K, K)$ (how a representative intertwines substitution with the deck action) and that an invariant homotopy forces a common twist on its two faces; the tautological class has the free-orbit twist $\delta = \mathrm{id}$. The main proof below does not need the twist calculus — the projections of Lemma 11.2 apply to invariant classes of every twist — but the twist point of view explains the stage-zero computations of Section 9 and we use it there.

12. Reachability and the two-sector induction

Definition 12.1. Let X be the set of invariant classes of the equivariant replacement of $\widetilde{C}$ (at level [2]) reachable from ι by finite chains of invariant homotopies.

If C is type-contractible then the vertex constant lies in X. We show it does not, by controlling how chains can leave the $\widetilde{M}$-sector. The subreplacement $R_K(J)$ of boxes supported in the middle copy $J = \widetilde{M} \cap \mathrm{cone}(J)$ is contained in both sector subreplacements (Lemma 11.1(2)); call an invariant class an *interface class* if it is a class of $R_K(J)$. On interface classes the retraction $\tilde{r}$ restricts to $\Phi \circ \pi$, and the descent of an interface class is a class of $[W, R(J/K)]$.

Lemma 12.2 (Projection of $\widetilde{M}$-chains)**.** *Let $e_0, \ldots, e_k$ be a chain of invariant classes and invariant homotopies in which every class and every homotopy is supported in the $\widetilde{M}$-sector or is an interface-to-interface homotopy supported in the cone sector without vertex support. Then applying $\tilde{r}$ to the $\widetilde{M}$-material and $\bar{\Phi} \circ \pi$ to the interface material yields a chain of invariant homotopies in $R_K(\square^2)$ from $\tilde{r}(e_0)$ to $\tilde{r}(e_k)$; in particular if $e_0 = \iota$ then $\mathrm{id}_W \simeq \tilde{r}(e_k)$ in $[W, R(W)]$.*

Proof. $\tilde{r}$ is totally defined on the $\widetilde{M}$-sector (Lemma 11.2(2)). A cone-sector homotopy between interface classes with no vertex support is a cell of the second cylinder to which π applies (Lemma 11.2(3)); its π-image is an equivariant homotopy of J-cells, and composing with Φ gives a homotopy of their $\tilde{r}$-images, since $\tilde{r}$ restricted to the interface is Φ (the first cylinder's retraction sends the middle copy of J to $\Phi(J) \subseteq \square^2$). The chains concatenate because the two projections agree on interface classes. $\square$

Lemma 12.3 (Sector pasting)**.** *Work in the equivariant replacement of $\widetilde{C}$ (Lemmas 2.2 and 11.1). Let b be a box datum for a box inclusion $j = (S \hookrightarrow \square^n) \mathbin{\hat{\otimes}} \delta^\varepsilon$ with distinguished coordinate w, let c_{top} denote the top cell of* $\mathrm{cod}(j)$*, and let $F := \mathrm{val}(b, c_{\mathrm{top}})$ be the fresh filler cell; by the relations (i) of Lemma 2.2 the faces of F at positions of* $\mathrm{dom}(j)$ *are the prescribed cells $b(c)$. Suppose b is* mixed*: its prescribed cells meet both sectors of Lemma 10.2, and its $w{=}0$-face $e := b(c_{\mathrm{top}} \cdot \delta^0_w)$ (the prescribed face at the $w{=}0$ endpoint) is an earlier-stage $\widetilde{M}$-sector cell. Then there exist an interface class G and filler cells F_M, F_c such that F_M lies in the $\widetilde{M}$-sector subreplacement with w-faces (e, G), and F_c lies in the cone-sector subreplacement with $w{=}0$-face G. Moreover a filler value is identified with an earlier-stage cell only at box-boundary positions $c \in$* $\mathrm{dom}(j)$*: the one face a filler creates — its face at the unique position outside* $\mathrm{dom}(j)$ *— is never equal to an earlier-stage cell.*

Proof. By Lemma 10.2 the prescribed cells $b(c)$, $c \in \mathrm{dom}(j)$, split into an $\widetilde{M}$-part (containing e) and a cone-part, meeting only in cells of the interface. First complete the interface data of b — the prescribed cells in which the two parts meet — to a full square G by filling inside the interface subreplacement $R_K(J)$: the required open boxes exist by inserting connection degeneracies of the interface edges, and their fillers exist by Lemma 2.2 applied to the subreplacement. Next form the box datum b_M with w=0-face e, w=1-face G, and sides the $\widetilde{M}$-side prescribed cells of b completed by connection degeneracies of the interface edges; its boundary is compatible because ∂G consists of the interface edges of b, to which the original sides of b already connect ∂e. Declare the missing face of b_M to be one of its *sides*; then the filler $F_M := \mathrm{val}(b_M, c_{\mathrm{top}})$ (in the $\widetilde{M}$-sector subreplacement, again by Lemma 2.2) has both w-faces prescribed, equal to e and G. The cone-side cell F_c is constructed symmetrically over the cone-part of b.

For the sector bookkeeping and the final claim we use the presentation of Lemma 2.2: identifications among filler values are generated exactly by (i) the box-boundary relations $\mathrm{val}(b, c) = b(c)$ for c in the open box, and (ii) the naturality relations $\mathrm{val}(b \circ \hat{\sigma}, c) = \mathrm{val}(b, \hat{\sigma} \circ c)$. Relations (ii) relate boxes to their reindexings, which have the same support, hence the same sector; relations (i) identify filler values with earlier cells only at positions c in the open box — never at the created face. Hence the sector of every cell above is well-defined, and a fresh created face is never identified with an earlier-stage cell, proving the last claim. □

Lemma 12.4 (First exit). *Suppose some chain from ι reaches a class with vertex support. Then some chain from ι of the restricted kind of Lemma 12.2 ends at an interface class j_0, and moreover $\tilde{r}(j_0) = R(\bar{\Phi}) \circ g$ for the invariant J-class $g := \pi(j_0)$.*

Proof (stage induction). Consider a chain from ι to a vertex-supported class and the first homotopy H in it whose target class fails the restriction of Lemma 12.2. We induct on the stage of H.

Stage 0. H is a cell of $\widetilde{C}$. If H is supported in $\widetilde{M}$ or is an interface-to-interface second-cylinder cell, it does not fail the restriction. The remaining stage-zero possibilities are second-cylinder cells with τ_2 vanishing on exactly one w-face: then that face is an interface class j_0 and the chain up to j_0 is restricted; the other face and all later material lie in the cone sector. This is the first-exit configuration, and $\tilde{r}(j_0) = \Phi \circ (j_0\text{-data}) = R(\bar{\Phi}) \circ \pi(j_0)$ by definition of the gluing. (Cells with vertex support have no $\widetilde{M}$-sector faces by Lemma 11.2(4), so they cannot be the first exit.)

Fresh fillers. Let $H = F = \mathrm{val}(b, c_{\mathrm{top}})$ be a fresh filler of a box datum b of earlier-stage cells, or a reindexing of one. If the prescribed cells of b are supported in one sector, so is F, and F is not a first exit (either it is restricted material, or its faces are all cone-sector and the earlier chain already exited). Suppose b is mixed. Apply the sector pasting lemma (Lemma 12.3) to b and its w=0-face e: it yields an interface class G and an $\widetilde{M}$-sector cell F_M with w-faces (e, G). Whatever the position of the missing face of b and whatever face the filler F creates, the chain that arrived at e through restricted material can therefore be continued by the single restricted step

F_M to the interface class G — the conclusion of the lemma — without any comparison between F and other fillers of the same box. One case requires comment: if the missing face of b is its w=0-face, then the w=0-face of F is the face F creates, and by the last clause of Lemma 12.3 a created face is never identified with an earlier-stage cell — not even through the coequalizer relations — so it cannot equal the earlier-stage restricted class e, and this case does not occur at a first exit.

Applying the inductive hypothesis to the earlier-stage constituents completes the induction: the first exit presents an interface class as claimed. □

Remark 12.5. The pasting step is where the subdivision is essential: in the unsubdivided cone a single cell can join the base to the vertex, the box does not decompose, and neither projection is defined on it. It is also the step whose full case analysis (positions of the missing face; reindexings; the equivariant compatibility of the pasted fillers) follows the proof of the nullity theorem (Theorem 6.3); the two proofs share the double induction over stages and dimensions, and we have kept here the geometry particular to C and refer to the proof of Theorem 6.3 for the generic bookkeeping.

Theorem 12.6 (Object-level separation)**.** *The subdivided mapping cone C of Construction 10.1 is test-contractible and not type-contractible. Consequently the two structures have different classes of aspherical objects, and $C \to \square^0$ lies in $\mathsf{W}_{\text{test}} \setminus \mathsf{W}_{\text{type}}$.*

Proof. Test-contractibility is Proposition 10.3. Suppose C is type-contractible. Then the vertex constant is reachable from ι, and it has vertex support, so by Lemma 12.4 there are an interface class j_0 and a restricted chain from ι to j_0. By Lemma 12.2, $\mathrm{id}_W \simeq \tilde{r}(j_0) = R(\bar{\Phi}) \circ g$ in $[W, R(W)]$, where $g = \pi(j_0) \in [W, R(J/K)]$. By the nullity theorem (Theorem 6.3) every such g is null-homotopic, so $\mathrm{id}_W \simeq R(\bar{\Phi}) \circ \mathrm{const} = \mathrm{const}$, making W type-contractible and contradicting Corollary 2.4.

For the consequences: $C \to \square^0$ is a test equivalence by test-contractibility; were it a type equivalence, C would be type-contractible. □

13. Machine cross-checks

Every finite claim of this paper is verified by exhaustive computation, in scripts accompanying the artifact repository of this series:

`https://github.com/r-shibusawa/cubical-strict-j`

(release `v1.16.0`, commit `31bf633`, archived at DOI 10.5281/zenodo.21927526; the scripts `scripts/collage_type*.py`, `scripts/homotopy_csp.py` and `scripts/object_census.py`). In particular: (i) the equivariance, self-duality and flow identities of Lemmas 3.2–3.3 are checked in the free De Morgan algebras on up to four generators, exactly, in the monotone-function representation; (ii) the classification of invariant cells (Proposition 5.4) is confirmed by the full census of the 168^3 level-two triples: 372 invariant classes (360 interior, 12 on the ends), the interior ones all with strictly invariant representatives — independently of the parity proofs given here; (iii) the collapse-locus closure properties (Lemma 4.2) are checked exhaustively at levels one and two; (iv) as negative controls, the census also verifies that no strict section of $\bar{\Phi}$ exists at either the cover or the quotient level, and that the four loop-twisted candidate squares are pairwise definitionally distinct before their common null-homotopy class is taken — the phenomena that motivated the parity analysis. A companion object-language development (the `LibJoinTC` module of the artifact's Lean 4 kernel) realizes the strictly K-invariant twisted squares inside cubical type theory itself and machine-checks their invariance definitionally; it played a heuristic role and is not used by the proofs.

For the object-level part, additionally: the twelve edge classes of W and their eight reversal-invariant members; the strict rigidity of the diagonal loop (Proposition 8.1, 168^2 states); the invariant square census and the connectivity of the free-homotopy graph (Proposition 8.2), including the explicit square $(u\wedge v,\ u\vee\neg v)$; the 72 strict endomorphism classes of W, the identity's strict-homotopy component of 3, and its disjointness from the 42 $\bar{\Phi}$-factoring classes (Proposition 9.2, computed by the satisfiability method of Proposition 9.1); and the freeness of the deck actions on cells. These checks are reproduced by `scripts/object_census.py` and `scripts/homotopy_csp.py` of the release named above.

14. Related work and outlook

That strict interval quotients break the correspondence between cubical models with reversals and spaces was established in [15] (completing a folklore expectation): both structures assign $K(\mathbb{Z}/2,1)$ to $\square^1/\langle\neg\rangle$, whose realization is a point. The question separated here — whether the two *cubical* structures agree with each other — is of a different nature, and the answer localizes the disagreement precisely: it is not reversal as such (the free family agrees), nor fixed points as such (the fixed-point-full family agrees), but the *mixture* of isotropy types, where the test structure builds a genuine Borel colimit while strict equivariance in the type structure is frozen by the parity lemmas.

The model structure comparison problem for cubical sites has been studied intensively in the Cartesian setting, where the equivariant refinement of [3] produces a type-theoretic model structure that *does* coincide with the test structure — making the De Morgan separation proved here a genuine dichotomy between sites rather than a defect of cubical methods. It is worth spelling out how the two situations differ. In the cartesian setting the obstruction addressed by [3] is already visible at the level of *objects*: symmetric quotients of cubes (the motivating example being the quotient of the square by the coordinate swap) fail to be contractible in the unrepaired type-theoretic structure, and the equivariant fibration condition is designed exactly so that such quotients contract, after which the repaired structure coincides with the test structure. On the De Morgan site the phenomenon is subtler: by [15], the Klein quotient W is non-contractible in *both* structures — indeed no isotropy quotient of the square separates the two localizers by object-level contractibility. The present paper therefore first constructs a *map-level* witness, $\bar{\Phi} : J/K \to W$, whose test-side equivalence and type-side nullity expose a separation invisible to the object-level criterion *within* the class of isotropy quotients; and then, leaving that class, an object-level witness after all (Theorem 12.6) — a finite complex, not an isotropy quotient, on which the two structures disagree about contractibility itself. The responsible mechanism (the parity rigidity of strict K-invariance) has no cartesian counterpart, since it turns on the reversal. For the De Morgan site, [12] constructs the type structure and [4] analyses (relative) elegance; the expectation of non-coincidence goes back to [13]. Our separation shows that on the De Morgan site no Quillen equivalence between the two structures can exist along the identity functor; whether type$(\widehat{\square_{\mathrm{DM}}})$ nonetheless presents spaces *abstractly* (via a different comparison) is exactly the question of computing localizations such as type(W), which the nullity mechanism makes approachable.

A recent result of Cavallo and Sattler [5] completes the picture from the type-theoretic side: reversal *as a type-theoretic operation* is innocuous. They prove that extending a cubical type theory with a self-dual interval theory by a reversal is conservative, and construct — by a twist construction, inside cubical sets *without* reversal — models of cubical type theory with reversal whose homotopy theory is that of topological spaces. Together with the equivariant repair of [3] in the cartesian setting, the landscape is now: reversal can be modelled compatibly with spaces ([5]), cartesian symmetric quotients can be repaired equivariantly ([3]), and yet the *standard* De Morgan cubical-type structure — the semantics in which CCHM type theory was originally interpreted — is separated from the spatial homotopy theory by the explicit witnesses of this paper, at the level of maps (Theorem 7.1) and of objects (Theorem 12.6). The three results are complementary and none subsumes another: ours concerns the standard structure on the standard site.

Three further directions. First, the *scope of the parity mechanism*: Lemmas 5.2–5.3 are statements about $\mathbb{Z}/2$-patterns in free De Morgan algebras; for other groups and dimensions the analogous character analysis should decide exactly which strict quotients separate the structures, refining the trichotomy of Corollary 7.3 into a complete classification. Second, the *Kleene and Boolean subvarieties*: the same questions on the sites of [14], where the fixed-line combinatorics differ. Third, *syntactic consequences*: the witness lives on the site of a proof assistant; the nullity theorem says definitional K-invariance forces homotopical triviality — a no-go complementary to those of [17], and a caution for internalizing strict quotients in cubical proof assistants.

Declarations.
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Data availability. The census and verification scripts supporting the finite claims are openly available in the repository named in Section 13, archived on Zenodo: release `v1.16.0`, commit `31bf633`, DOI 10.5281/zenodo.21927526.
Competing interests. The author declares no competing interests.

Appendix A. Proof of (W2)

We prove $\mathrm{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ using only (Collage) and elementary covering arguments; the part of (L) that we need is reproved en route. With this appendix and Proposition 2.3, no statement of [15] enters the proofs of Theorems 7.1 and 12.6: the collage formula (Collage) is used directly from [1, 18].

The stratification. A nonempty cell (φ, ψ) of $\square^2$ is fixed by sw iff it is *diagonal* $(\psi = \varphi)$, and by g iff it is *antidiagonal* $(\psi = \neg\varphi)$; nb fixes no nonempty cell, since no element of a free De Morgan algebra equals its own negation. The diagonal cells form a subpresheaf isomorphic to $\square^1$ (parametrized by φ), on which the residual action of $K/\langle \mathrm{sw} \rangle$ is the reversal $\varphi \mapsto \neg\varphi$; likewise for the antidiagonal. Hence the images $\bar{D}$, $\bar{A}$ of the two strata in W are each isomorphic to L, and they are disjoint (a common cell would satisfy $\varphi = \neg\varphi$). A cell of W outside $\bar{D} \sqcup \bar{A}$ has trivial stabilizer: a nontrivial stabilizing element would be sw or g (placing the cell on a stratum) or nb (impossible).

The collage. $S := \bar{D} \sqcup \bar{A}$ is a subpresheaf of W, hence $\mathrm{el}(S)$ is a sieve in $\mathrm{el}(W)$, with complementary cosieve U the full subcategory of stabilizer-free cells. By (Collage), $N\mathrm{el}(W)$ is the homotopy colimit of a zig-zag built from $N\mathrm{el}(S)$, NU, and the nerves of the two comma parts $\widehat{W}_D$, $\widehat{W}_A$ (pairs consisting of a stratum cell, a cosieve cell, and a connecting map in $\mathrm{el}(W)$; the comma part splits according to the stratum of the source since $\bar{D}$ and $\bar{A}$ are disjoint).

The sieve pieces. We claim $\mathrm{el}(\mathrm{L}) \simeq B\mathbb{Z}/2$, which is the part of (L) used here. The category $\mathrm{el}(\square^1)$ has a terminal object (the generating cell x), hence is contractible; the reversal acts freely on the nonempty cells of $\square^1$ (no element of a free De Morgan algebra equals its own negation), so the quotient functor $\mathrm{el}(\square^1) \to \mathrm{el}(\mathrm{L})$ is a covering of categories with free transitive $\mathbb{Z}/2$-action on each fiber, compatibly with all structure maps. Hence $N\mathrm{el}(\mathrm{L}) \cong N\mathrm{el}(\square^1)/(\mathbb{Z}/2) \simeq B\mathbb{Z}/2$. Consequently $N\mathrm{el}(S) = N\mathrm{el}(\bar{D}) \sqcup N\mathrm{el}(\bar{A}) \simeq B\mathbb{Z}/2 \sqcup B\mathbb{Z}/2$.

The cosieve piece. Let $F \subseteq \mathrm{el}(\square^2)$ be the full subcategory of cells with trivial stabilizer. The generating cell $(x, y) \in \square^2([2])$ lies in F (it is on no stratum), and every cell of $\square^2$ is a unique substitution instance of it, so F

has a terminal object and NF is contractible. The group K acts freely on F, and the quotient functor $F \to U$ is a covering of categories: each U-cell has exactly $|K| = 4$ preimages, permuted freely and transitively by K, compatibly with all structure maps (the quotient $\square^2 \to W$ is levelwise the orbit map, and orbits of stabilizer-free cells are free). Hence $NU \cong NF/K \simeq BK$, the standard Borel model of the classifying space of $K = \mathbb{Z}/2 \times \mathbb{Z}/2$.

The comma pieces. Consider $\widehat{W}_D$. Lifting along the two coverings just described, $\widehat{W}_D$ is isomorphic to the quotient by K of the category $\mathcal{A}_D$ of triples (a cell d of the diagonal copy of $\square^1$ inside $\square^2$, a cell $u \in F$, a map $d \to u$ in $\mathrm{el}(\square^2)$), where K acts diagonally; the action on $\mathcal{A}_D$ is free because it is already free on the F-component. The projection $\mathcal{A}_D \to D_F$ onto the first component (where $D_F \cong \mathrm{el}(\square^1)$ is the diagonal copy) is a Grothendieck fibration whose fiber over d is the coslice $(d \downarrow F)$; this coslice has a terminal object, namely the generating cell together with the canonical map $d \to (x, y)$, so each fiber is contractible. Since the base $D_F \cong \mathrm{el}(\square^1)$ has a terminal object as well (hence is contractible), Quillen's Theorem A gives that $N\mathcal{A}_D$ is contractible. Hence $N\widehat{W}_D \cong N\mathcal{A}_D/K \simeq BK$, and likewise $N\widehat{W}_A \simeq BK$.

Evaluating the colimit. The collage formula presents $N\mathrm{el}(W)$ as the homotopy colimit of the zig-zag

$$N\mathrm{el}(\bar{D}) \longleftarrow N\widehat{W}_D \longrightarrow NU \longleftarrow N\widehat{W}_A \longrightarrow N\mathrm{el}(\bar{A}),$$

i.e. of $B\mathbb{Z}/2 \leftarrow BK \rightarrow BK \leftarrow BK \rightarrow B\mathbb{Z}/2$, where the outer legs are induced by the projections to the strata and the inner legs by the projections to the cosieve. The inner legs are weak equivalences: on the free covers they are the contractions $N\mathcal{A}_D \to NF$ computed above, and both sides are free K-quotients of contractible complexes, so each inner leg is a map $BK \to BK$ inducing the identity on $\pi_1 = K$. Collapsing the inner legs, the colimit reduces to the homotopy pushout

$$B\mathbb{Z}/2 \longleftarrow BK \longrightarrow B\mathbb{Z}/2.$$

On fundamental groups the left leg is induced by the residual action of $K/\langle \mathrm{sw} \rangle$ on $\bar{D}$ (a comma object over a diagonal cell remembers the stabilizer $\langle \mathrm{sw} \rangle$, which acts trivially on the stratum), i.e. by the projection $K \to K/\langle \mathrm{sw} \rangle \cong \mathbb{Z}/2$; likewise the right leg is induced by $K \to K/\langle g \rangle$. Since $K = \langle \mathrm{sw} \rangle \times \langle g \rangle$, the two legs are the two projections of $B\mathbb{Z}/2 \times B\mathbb{Z}/2 \simeq BK$ onto its factors, and the homotopy pushout of the two projections of a product is by definition the join. Hence

$$N\mathrm{el}(W) \;\simeq\; B\mathbb{Z}/2 \star B\mathbb{Z}/2.$$

The join of two connected complexes is simply connected, and the reduced homology of a join gives $\tilde{H}_i(B\mathbb{Z}/2 \star B\mathbb{Z}/2; \mathbb{Z}) \cong \bigoplus_{p+q=i-1} \tilde{H}_p(B\mathbb{Z}/2) \otimes \tilde{H}_q(B\mathbb{Z}/2) \oplus \bigoplus_{p+q=i-2} \mathrm{Tor}(\tilde{H}_p B\mathbb{Z}/2, \tilde{H}_q B\mathbb{Z}/2)$, whose first nonvanishing group is the tensor contribution $\tilde{H}_3(B\mathbb{Z}/2 \star B\mathbb{Z}/2;\mathbb{Z}) \cong \tilde{H}_1(B\mathbb{Z}/2) \otimes \tilde{H}_1(B\mathbb{Z}/2) \cong \mathbb{Z}/2$ (the Tor summands vanish in degrees ≤ 3). In particular $\mathrm{el}(W)$ is not weakly contractible. This proves (W2). $\square$

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